

Email Sending Service

1. Project Title

Email Sending Service

2. Project Overview

An Email Sending Service is a backend application that sends emails automatically to users. It can be integrated with websites, mobile apps, and business systems for notifications, verification, password resets, newsletters, and promotional messages.

Examples:

Welcome Email after Registration

OTP Verification Email

Password Reset Email

Order Confirmation Email

Newsletter Email

3. Problem Statement

Many applications need to send emails automatically. Managing email delivery manually is inefficient and time-consuming.

An Email Sending Service provides:

Automated email delivery

Reliable communication

User verification

Notification management

4. Objectives

Send emails automatically.

Verify user accounts using OTP.

Handle password reset requests.

Send bulk emails and newsletters.

Track email delivery status.

5. Scope of the Project

The system allows:

User Communication

Welcome Emails

Account Verification Emails

Password Reset Emails

Marketing Emails

Newsletters

Promotional Emails

Security Emails

OTP Verification

Login Alerts

Tracking

Email Delivery Status

Email Logs

6. System Architecture

Plain text

User/Application

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Email Sending Service API



SMTP Server / Email Provider



Recipient Mailbox

Components

Client Application

Sends email requests.

Backend API

Processes requests.

Generates email content.

Sends emails.

Email Provider

Examples:

Gmail SMTP

SendGrid

Mailgun

Recipient

Receives emails in their inbox.

7. Functional Requirements

Welcome Email Module

When a user registers:

Input:

Name

Email

Output:

Welcome Email

Example:

Plain text

Subject: Welcome

Hello John,

Thank you for joining our platform.

OTP Verification Module

When a user signs up:

Input:

Email Address

Output:

OTP Email

Example:

Plain text

Your OTP is: 548921

Password Reset Module

When a user forgets their password:

Input:

Registered Email

Output:

Password Reset Link

Example:

Plain text

<https://example.com/reset-password/token123>

Newsletter Module

Send emails to multiple users.

Example:

Plain text

Monthly Updates

Latest Features

Offers and Announcements

Bulk Email Module

Send emails to many recipients simultaneously.

Use Cases:

Event Notifications

Product Launches

Marketing Campaigns

8. Non-Functional Requirements

Security

Email validation

Secure SMTP authentication

OTP expiration

Performance

Fast email delivery

Reliability

Accurate email sending

Scalability

Handle thousands of emails

9. Database Design

Users Table

Field

Type

user_id

Integer

name

Varchar

email

Varchar

created_at

DateTime

Email Logs Table

Field

Type

email_id

Integer

recipient

Varchar

subject

Varchar

status

Varchar

sent_time

DateTime

OTP Table

Field

Type

otp_id

Integer

email

Varchar

otp_code

Integer

expires_at

DateTime

10. API Endpoints

Send Email

Http

POST /api/send-email

Request:

JSON

```
{  
  "to": "user@example.com",  
  "subject": "Welcome",  
  "message": "Welcome to our platform"  
}
```

Send OTP

Http

POST /api/send-otp

Request:

JSON

```
{  
  "email": "user@example.com"  
}
```

Verify OTP

Http

POST /api/verify-otp

Request:

JSON

```
{  
  "email": "user@example.com",  
  "otp": "548921"  
}
```

Send Newsletter

Http

POST /api/newsletter

11. Technology Stack

Backend

Node.js

Framework

Express.js

Email Library

Nodemailer

Database

MongoDB

Testing Tool

Postman

12. Project Workflow

Step 1

User performs an action.

Example:

Registration

Password Reset

Step 2

Application sends a request to the Email Service API.

Step 3

Backend validates the request.

Step 4

Email content is generated.

Step 5

Email is sent through the SMTP provider.

Step 6

Email status is recorded in the database.

Step 7

User receives the email.

13. Sample API Request

Send Welcome Email

Request:

JSON

```
{  
  "to": "john@example.com",  
  "subject": "Welcome",  
  "message": "Thank you for registering."  
}
```

Response

JSON

```
{  
  "message": "Email sent successfully"  
}
```

14. Advantages

Automated communication

Improves user experience

Supports account verification

Secure password recovery

Useful for marketing campaigns

Easy integration with other applications

15. Future Enhancements

Email Templates

Create professional HTML email designs.

Scheduled Emails

Send emails at a specific date and time.

Email Analytics

Track:

Delivery Rate

Open Rate

Click Rate

Attachment Support

Send:

PDFs

Images

Documents

Queue System

Use message queues for large-scale email processing.

Multi-Provider Support

Switch automatically between email providers if one fails.

16. Project Conclusion

The Email Sending Service is a backend application that automates email communication between applications and users. It supports notifications, OTP verification, password recovery, newsletters, and bulk messaging. The project demonstrates important backend development concepts such as API creation, SMTP integration, database management, authentication, logging, and scalability, making it an excellent backend development project for academic submissions and professional portfolios.

